Muhammad Ibtasaam Amjad

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Education

National University of Computer and Emerging Sciences, Islamabad Campus **Aug. 2024 – June 2028**
Bachelor of Science in Computer Science (CGPA: 3.86) *Islamabad, Pakistan*

Punjab College, Ayub Park Campus **2022 – 2024**
Intermediate in Computer Science — Federal Board; 982/1100 marks *Rawalpindi, Pakistan*

Fauji Foundation College for Boys, New Lalazar **2020 – 2022**
Matriculation, Science Group (Biology) — Federal Board; 1074/1100 marks *Rawalpindi, Pakistan*

Certifications

Ordered Data Structures **July 2025**
University of Illinois Urbana-Champaign

Microsoft Office Specialist: Word Associate (Word and Word 2019) **December 2022**
Microsoft

Graphic Designer **August 2021**
Digital Youth Pakistan

Experience

Verteon **July 2026 – Present**
Chief Technology Officer *Islamabad, Pakistan*

- Serve as CTO of a startup incubated in the 8th Hatch at the National Science and Technology Park (NSTP).
- Architected an offline posture-correction desktop application combining an ESP32 pressure-sensing cushion, Rust/Tauri sensor telemetry, SQLite session data, and React analytics.

Parallel Computing Networks Lab **July 2026 – Present**
Intern *Islamabad, Pakistan*

- Build data ingestion, extraction, and analysis workflows at enterprise scale using Apache Spark and Kafka.
- Research NAaaS/NewsNet news knowledge-graph construction, including entity extraction and resolution, relation extraction, article linking, and graph-community analysis; benchmarked DeBERTa-v3 NER against a spaCy baseline.

Roman Urdu–English Language Identification Research **October 2026 – Present**
Research collaborator

- Collaborating on ongoing token-level language identification research for Roman Urdu–English code-mixed text, focusing on ambiguous words, spelling variation, and category-wise evaluation.

FAST-NUCES, Islamabad Campus **August 2026 – Present**
Teaching Assistant — Programming Fundamentals, CS 2026 cohort *Islamabad, Pakistan*

- Assist the CS 2026 cohort with Programming Fundamentals concepts and coursework.

FAST-NUCES, Islamabad Campus **August 2026 – Present**
Teaching Assistant — Arts and Humanities, SE 2026 cohort *Islamabad, Pakistan*

- Support Arts and Humanities coursework for the SE 2026 cohort.

FAST-NUCES, Islamabad Campus **August 2026 – Present**
Teaching Assistant — Linear Algebra, CS 2025 cohort *Islamabad, Pakistan*

- Guide the CS 2025 cohort through Linear Algebra coursework and problem solving.

FAST, Islamabad Campus **January 2026 – June 2026**
Teaching Assistant — Object Oriented Programming, CS 2025 cohort *Islamabad, Pakistan*

- Designed quizzes and assignments, evaluated the semester project, and gave students feedback on object-oriented design.

FAST-NUCES, Islamabad Campus **August 2025 – January 2026**
Teaching Assistant — Programming Fundamentals, CS 2025 cohort *Islamabad, Pakistan*

- Created and assessed foundational programming work, guiding students through the semester project.

FAST-NUCES, Islamabad Campus **January 2026 – June 2026**
Teaching Assistant — Expository Writing, CS 2025 cohort *Islamabad, Pakistan*

- Evaluated writing assignments and quizzes for the CS 2025 cohort.

FAST Artificial Intelligence Society **September 2025 – June 2026**
Technical Projects Officer *Islamabad, Pakistan*

- Engineered advanced automated workflows using LangChain and n8n, seamlessly integrating code-based AI environments with visual node-based systems.
- Deployed these scalable automation solutions for university-wide use, bridging the gap between development and institutional implementation.

FAST Computing Society **September 2025 – June 2026**
Social Media Officer *Islamabad, Pakistan*

- Directed the creative flow of content published by the society.

Desirotech **July 2023 – June 2026**
Freelance Graphic Designer *Rawalpindi, Pakistan*

- Developed the software house's brand identity and visual design language, creating a consistent look for its marketing and digital materials.

Projects

The Odyssey: Swarm of Multi-Agent Systems | *Python, FastAPI, MCP, Redis, Docker, n8n, TypeScript, WhatsApp Evolution API* **July 2026**

- Architected Iris, a WhatsApp-facing ReAct orchestrator that delegates through MCP to specialist systems for knowledge retrieval (Hikma/Maktaba), long-form writing (Musannif), Google Workspace (Google MAS), and software engineering (Coding MAS).
- Implemented per-sender memory, deferred task completion, Redis-backed event telemetry, and a live swarm dashboard; documented end-to-end WhatsApp flows for Hikma and Musannif.

Google MAS: Workspace Agent System | *Python, Google Workspace APIs, OAuth, MCP, Docker* **July 2026**

- Built multi-account Google Workspace agents for Gmail, Calendar, Drive, Classroom, Tasks, and Contacts, exposing task-level tools to The Odyssey through an HTTP MCP service.

Coding MAS: Multi-Agent Software Engineering | *Python, MCP, Git, automated tests, Docker* **July 2026**

- Built a specialist coding workflow for The Odyssey with Apollo/Momus planning and critique, isolated feature-branch execution by agy Forge, test feedback, an action ledger, and completion callbacks.

Wisdom / Kybernetes: Obsidian Knowledge System | *Obsidian, Python, MCP, Markdown* **2026**

- Built and maintained a versioned Obsidian knowledge base with 380+ content notes, linked table-of-contents hubs, and consistent tagging conventions.
- Developed Python MCP tools for note management, search, and agent workflows, and integrated Musannif to generate and classify long-form notes into the vault.

Musannif: Multi-Agent Long-Form Writer | *Python, Textual, Typer, NetworkX, asyncio, MCP, Anthropic/Mistral APIs* **2026**

- Built a terminal application that turns a topic into an editable hierarchical outline and schedules sections on a dependency graph for concurrent generation by nine domain specialists.
- Implemented context handoff through section summaries and optional full-text retrieval, assembled the result as Markdown, and routed finished notes into the Kybernetes vault.

Maktaba: AI-Assisted Knowledge Library | *Next.js 15, React 19, TypeScript, D3, Supabase, MiniLM embeddings* **2026**

- Built a knowledge library from the Kybernetes vault, publishing 370 linked notes and roughly 896,000 words through an incremental Markdown pipeline.
- Generated a 4,196-section semantic index and interactive graph, with Hikma, an AI research companion that can search and read note sections.

AI Agent Security: Multi-Step Tool Attacks | *Python, Kaggle, aicomp_sdk* **2026**

- Developed red-team attack algorithms for the OpenAI/Google/IEEE Kaggle challenge, probing a sandboxed tool-using agent and returning replayable security-failure traces.
- Tested seven prompt templates and selected strategies using replay-validated success and latency, with candidate deduplication and evaluation-budget controls.
- Reached a peak public leaderboard rank of 40th worldwide.

Verteon: Offline Posture-Correction System | *Tauri 2, Rust, React 19, TypeScript, SQLite, ESP32, BLE, USB serial, Python, MediaPipe, OpenCV* **July 2026**

- Architected an offline desktop system pairing a four-sensor ESP32 cushion with BLE/USB telemetry, a Rust posture-state engine, SQLite session tracking, and React analytics.
- Built webcam-guided calibration with a Python MediaPipe/OpenCV sidecar and a Windows CI workflow that packages and smoke-tests a live installer.

TaxNet: Graph-Based Tax Intelligence Prototype | *Next.js 16, React 19, TypeScript, FastAPI, Neo4j GDS* **June 2026**

- Built a graph-based tax intelligence prototype in 48 hours, including an entity-resolution pipeline with phonetic blocking, sentence-transformer matching, IDF-weighted address similarity, and logistic-regression fusion; achieved approximately 0.90 F1 on more than 110,000 records at near-linear runtime.
- Applied PageRank, community detection, and cycle analysis for graph-based pattern detection; scored entities with Shapley decomposition and Positive-Unlabeled learning (Elkan-Noto).
- Developed Next.js/TypeScript dashboards for entity detail, graph exploration, and analyst queries backed by FastAPI and Neo4j services.

NAaaS / NewsNet: News Knowledge-Graph Research | *Python, spaCy, DeBERTa-v3, sentence-transformers, FAISS, RapidFuzz, Leiden* **July 2026**

- Researching entity extraction and resolution for a news knowledge graph, with an offline pipeline for relation extraction, embeddings, article linking, and graph communities.
- Evaluated fine-tuned DeBERTa-v3 NER at 93.68% micro F1 on 8,262 OntoNotes 5.0 test sentences versus an 87.19% spaCy baseline, and developed audited ECB+/CoNLL-2012 resolution benchmarks.

Pokayoke: Pre-Submission Financial Data QA | *Python, FastAPI, Google ADK, MCP, pandas, scikit-learn, Gemini, React, TypeScript, GCP Cloud Run* **October 2026**

- Co-built a first-place national AI hackathon platform that holds suspect financial CSV records before database release, presents evidence-backed corrections, and gates acceptance through analyst review.
- Implemented FastAPI/Google ADK agents with MCP tools, statistical and historical checks, versioned review, and idempotent release; integrated a branded React console and deployed demo on Cloud Run.

GridGuard: Electricity-Theft Inspection Dashboard | *Python, FastAPI, NumPy, scikit-learn, JavaScript, SVG, Cytoscape, Anthropic Claude* **July 2026**

- Built a Pakistan-wide inspection dashboard ranking 7,442 synthetic meters across 11 DISCOs through interpretable meter anomalies and transformer energy-balance checks.
- Added geographic drill-down, explainable risk scores, and optional AI-assisted field reporting and queries.

EduGap: Curriculum–Job-Market Gap Identifier | *Python, FastAPI, React, Vite, Gemini, ChromaDB, sentence-transformers, SQLite, Playwright* **June 2026**

- Co-developed a curriculum-to-job-market gap analyzer that extracts skills from course PDFs, compares them with job postings, and distinguishes missing, unassessed, and well-covered skills.
- Integrated a four-stage analysis pipeline with a Socratic tutor, shared agent memory, concept graphs, and a React dashboard.

Baghdad Reimagined: Classical-AI City Simulator | *Python, FastAPI, Pydantic, NumPy, React 19, TypeScript, Vite* **2026**

- Co-developed an interactive 112-sector Baghdad city-planning simulator; implemented Min-Conflicts zoning, greedy and genetic-algorithm ambulance placement, and flood-aware emergency rerouting.
- Integrated classical-AI planning services with a React interface for exploring road, emergency-response, and city-layout scenarios.

Proxenos: B2B Trade Orchestration Platform | *Java 21, Spring Boot, JavaFX, PostgreSQL, Python, FastAPI, NEAT, Solidity, Web3j* **2026**

- Co-developed a B2B trade platform spanning buyer and seller workflows, shipment state transitions, escrow payments, and dispute resolution across a Spring Boot service and PostgreSQL database.
- Helped integrate Solidity/Hardhat escrow contracts with Java payment orchestration and a Python routing service that filters feasible paths by budget and time, then scores them with NEAT or a heuristic fallback.

Chrono Rift: Multi-Process Tactical Game | *C++17, POSIX shared memory, pthreads, semaphores, signals, SFML, Docker* **Operating Systems semester project, Spring 2026**

- Co-developed a tactical game split into Arbiter, player-input, and NPC strategy processes, with shared-memory synchronization, FIFO stamina-based turns, and deadlock detection.
- Implemented graphical player controls, player-thread input, cross-process signalling, Docker launch support, and shutdown integration.

Islamagood: Islamabad Smart-City Simulator | *C++, WinAPI, ConsoleGraphicsExtension, custom graphs, hash tables, heaps, trees, queues* **Data Structures semester project**

- Co-developed a console city simulator with custom CGX visuals, animated transit, population and traffic heatmaps, and transport, healthcare, housing, and education views.
- Implemented graph-backed city navigation and traffic-weighted shortest-path routing alongside hand-built hash, heap, tree, and queue structures for city records and services.

Super Mario Bros in MASM | *x86 MASM, Irvine32, Windows console, WinMM* **Computer Organization and Assembly Language semester project**

- Built a multilevel Mario-inspired platformer in x86 MASM with tile-map rendering, collision and gravity, enemies and boss stages, a secret area, sound effects, and persistent high scores.

ConsoleGraphicsExtension (CGX) | *C++, WinAPI, GDI, ANSI true color, WinMM* **2025–2026**

- Built a Windows console graphics library with GDI pixel and ANSI true-color rendering paths, software framebuffers, sprite/text/geometry drawing, and keyboard/mouse input.
- Packaged reusable UI controls, animation and audio features with a Visual Studio sample application and interactive showcase.

DocuMind: AI-Driven Content Extraction | *n8n, LlamaParse, Pinecone, Gemini 1.5 Pro, Next.js* **October 2025**

- Architected a dual-stage RAG pipeline using LlamaParse and n8n to ingest complex PDFs, implementing custom regex-based page stamping to maintain 100% structural integrity across vector embeddings.
- Engineered an automated vector ingestion system in Pinecone with namespace-level flushing and metadata injection, enabling high-precision AI citations back to specific source page numbers.
- Developed a Next.js workspace featuring a 'Semantic Sync' engine that programmatically navigates an integrated PDF viewer to highlight specific document segments based on LLM-extracted JSON metadata.

Sonic Classic Heroes | *C++, SFML 2.6.2, Visual Studio* **Second semester (2025)**

- Co-built a four-level platformer with randomized chunk-based stages, switchable characters, enemy and boss encounters, collision and projectile systems, save/load, and a top-10 leaderboard.

Gaianox: Procedural 2D World Prototype | *C++17, SFML* **2025**

- Built a C++/SFML procedural-world prototype with nine-chunk streaming, binary saves, camera tracking, and animated movement.

Bresenham Line-Drawing Visualizer | *C++, SFML 2.6.2* **2023**

- Recreated Bresenham's integer line-drawing algorithm in an interactive 120-by-80 pixel-grid visualizer with mouse-selected endpoints.

Buzz Bomberz | *C++, SFML* **December 2024**

- Recreated Intellivision Buzz Bombers in C++/SFML with escalating arcade levels and scoring.

Space Invaders | *Python, Pygame* **2023**

- Built a Space Invaders-style shooter in Python/Pygame with alien waves, projectiles, and power-ups.

Socket Chess | *Python, Pygame, sockets, ngrok* **2023**

- Built networked two-player chess in Python/Pygame using sockets and ngrok for remote play.

Two Birds One Stone | *Python, Pygame* **2023**

- Built an original slingshot-and-bird arcade game in Python/Pygame with projectile motion and environmental hazards.

2D Voxel Engine | *Python, Pygame* **2023**

- Built an interactive 2D voxel-terrain prototype in Python/Pygame.

2D Voxel Engine V2 | *Python, Pygame* **2023**

- Expanded the 2D voxel engine in Python/Pygame with generated terrain, enemies, and combat.

Flappy Bird Recreation | *Python, Pygame* **2023**

- Recreated Flappy Bird in Python/Pygame with keyboard-controlled flight and obstacle navigation.

Pong Recreation | *Python, Pygame* **2023**

- Recreated two-player Pong in Python/Pygame with paddle controls and ball collisions.

Snake Recreation | *Python, Pygame* **2023**

- Recreated Snake in Python/Pygame with grid movement, food collection, and scoring.

Save the Ball | *Python, Pygame* **2023**

- Built a Chrome Dino-inspired survival game in Python/Pygame, guiding a bouncing ball past hazards.

Chess Board Prototype | *C++, SFML 2.6.2* **2023**

- Built an 8-by-8 chess-board prototype in C++/SFML with textured pieces and basic move and capture validation.

Bean Collector | *Python, Pygame* **2023**

- Built a timed coffee-bean collection arcade game in Python/Pygame.

NU-Hub | *HTML, CSS, JavaScript* **July 2025**

- Created a small resource hub for incoming NU students with links to past papers, courses, faculty information, societies, and practical tips.

Atari Breakout Recreation | *Python, Pygame* **2023**

- Built a Breakout-style arcade game in Python/Pygame with keyboard-controlled paddle, ball and brick collisions, and randomized brick layouts.

Technical Skills

Languages: Python, Java, C, C++, Rust, JavaScript, TypeScript, SQL, Solidity, x86 Assembly (MASM/NASM)
Frontend and Desktop: HTML/CSS, React, Next.js, Vite, Tauri, JavaFX, SFML, Pygame, Textual
Backend and Agent Systems: FastAPI, Spring Boot, MCP, Google ADK, LangChain, n8n, Web3j
Data Pipelines and ML: Apache Spark, Apache Kafka, spaCy, scikit-learn, sentence-transformers, MediaPipe, OpenCV
Databases and Search: PostgreSQL, SQLite, Supabase, Neo4j GDS, Redis, ChromaDB, FAISS, Pinecone
Development and Testing: Git, GitHub, VS Code, Visual Studio, JetBrains IDEs, pytest, Playwright
Environments, CI/CD and Deployment: Linux, Docker, Docker Compose, GitHub Actions, Google Cloud Run, Vercel

Leadership & Extracurricular

FAST Computing Society **August 2026 – Present**
Tech Wing Head *FAST-NUCES, Islamabad*

- Coordinate the Computing Society's technical activities and projects.

FAST Computing Society — Buddy Wing **August 2026 – Present**
Chairperson *FAST-NUCES, Islamabad*

- Coordinate peer mentorship and teaching between senior and junior students.

Meri Kahani / FAST Outreach and Engagement Society **August 2026 – Present**
Vice President, Internals *FAST-NUCES, Islamabad*

- Coordinate internal society activities and team operations.

Society for Innovation in Mathematics and Theoretical Sciences (SIMT) **August 2026 – Present**
Vice President, Externals *FAST-NUCES, Islamabad*

- Coordinate external outreach for the mathematics and theoretical sciences society.

NaSCon 2026 — Speed UML **2026**
Head of Event

- Led organization of the Speed UML event at NaSCon 2026.

NaSCon 2026 — Speed Programming **2026**
Vice Head of Event

- Supported organization of the Speed Programming event as vice head.

Honors and Awards

Dean's List: Fall 2024, Spring 2025, Fall 2025, and Spring 2026.
1st Place — National AI Hackathon (October 2026): Pokayoke, a pre-submission financial data-quality platform.
Silver Medalist (April 2025): Earned the second-highest SGPA in the Fall 2024 semester of BS Computer Science.
1st Place — Hackathon (July 2026): GridGuard electricity-theft inspection dashboard.
2nd Place — Islamabad Regional Hackathon (June 2026): EduGap curriculum-to-job-market gap identifier.
4th Place — First Hackathon (October 2025): DocuMind; the first hackathon I entered.
5th Place — Hackathon (June 2026): TaxNet graph-based tax intelligence prototype.
International Kangaroo Mathematics Competition (IKMC) (2022): Silver Medalist.
ICPC National Preliminaries (November 2025): Placed 1st at FAST Islamabad Campus and 21st in the national preliminaries.
WWF Spellethon: Multiple-time winner.
School Badminton Champion: Won the school championship multiple times.